Tony Lin

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Education

University of Virginia, School of Engineering and Applied Science — Charlottesville, VA
GPA: 3.99/4.0 — B.S. Computer Science and Math — *Aug. 2024 – May 2028*
- **Relevant Coursework:** Natural Language Processing, Probability, Mathematical Statistics, Data Structures & Algorithms

Thomas Jefferson High School for Science and Technology — Alexandria, VA
GPA: 4.394/4.0 — *Aug. 2020 – June 2024*
- **Award:** 2024 AIME Qualifier (American Invitational Mathematics Examination)

Experience

Enterprise AI Intern — May 2026 – Aug. 2026
Humana — *Remote*
- **Designed and built** an evaluation framework benchmarking Humana's HEDIS engine (a **LangGraph** pipeline extracting compliance evidence from clinical documents) against clinician-reviewed ground truth across **8 quality measures**
- **Developed** automated evals – recall/precision, **LLM-as-a-judge** scoring, confidence calibration (**ECE, Brier score**) – gating prompt and model changes before production promotion
- **Raised** extraction precision from **76% to 97%** for the OMW measure model, driving prompt refinement through error analysis and nurse feedback

AI/ML Intern — May 2025 – Aug. 2025
National Institute of Standards and Technology — *Hybrid*
- **Engineered** a RAG pipeline (LangChain, ChromaDB) benchmarking BM25, cosine similarity, and ensemble retrieval – **boosting** accuracy 12% and **cutting** latency 30% across 4k+ knowledge base articles
- **Automated** ingestion of 10k+ multimodal documents via Docling and VLMs, **reducing** manual processing effort by 80%

Machine Learning Evaluation Intern — Nov. 2024 – May 2025
ML@UVA — *Remote*
- **Partnered** with 6 teammates and client LMI to benchmark GraphRAG (BLEU, LLM-as-a-judge, cost, latency) – findings **adopted** for client pilot trials
- **Benchmarked** RAPTOR, CRAG, Self-RAG, and LightRAG against vanilla RAG on a legal dataset – found full GraphRAG's 10x cost and 100x latency outweighed its marginal gains, while LightRAG delivered the best quality/cost tradeoff

Projects

TDILEARNED | *Next.js, React Flow, FastAPI, LangGraph, Supabase* — Aug. 2026 – Present
- **Built** a multi-agent research pipeline (planner, parallel workers, verifier, synthesizer) that turns a topic into an interactive mindmap, sourcing DuckDuckGo, Wikipedia, Wikimedia, and OpenStreetMap
- **Streamed** live agent activity and chat over **SSE** from a **FastAPI** backend, validating output against **Pydantic** schemas with multi-provider failover
- **Precomputed** 1,000+ topic hubs with two-tier caching; deployed on **Vercel** + **Azure** behind a **Cloudflare** edge proxy

Dime | *FastAPI, Next.js, DuckDB, Gemini, Azure, Vercel* — May 2025 – Present
- **Shipped** a live NBA analytics agent (**Vercel** + **Azure**): plain-English Q&A over a self-hosted **DuckDB** warehouse spanning 2009-10–2025-26, with RAPTOR ratings back to 1976-77
- **Designed and built** the agent harness: plan-execute-verify loop over **101 tools** with execution policies, checkpoints, and a run ledger; multi-model routing (**Gemini**, **NVIDIA NIM**) behind one interface
- **Built** the data pipeline end to end: backfilled 330k+ box-score rows (2015-16–2024-25) with a coverage registry, so the agent declines out-of-range questions instead of hallucinating

Snaek's Value List | *Supabase, Azure Web App, Discord.js, Netlify, Cloudflare* — Feb. 2026 – Apr. 2026
- **Built and shipped** a value-tracking platform (trade-ad posting, calculator) for a Counter Blox (Roblox) trading community, reaching **1,000 monthly users** and replacing a manual spreadsheet as the source of truth
- **Integrated** a VLM-based inventory parser (Qwen3.6-27B via Groq) letting users upload a screenshot of their inventory and auto-populate item values, cutting manual entry and reaching **90% parsing accuracy**

Skills

Programming Languages: Python, TypeScript, JavaScript, Java
Frameworks & Libraries: React, Next.js, React Flow, Tailwind CSS, LangChain, LangGraph, Agno, FastAPI, Pydantic
Platforms & Tools: Git, GitHub, Supabase, PostgreSQL, Cloudflare Workers, Azure, Docker, Databricks, Langfuse, VS Code, OpenCode, Claude Code